



**Indian Institute of Space Science and
Technology (IIST)**

Department of Avionics
5G Use Case Lab

Installation Manual

OMNeT++, INET Framework and Simu5G

Installation using Windows 11, WSL2 and opp_env

Contents

1	Introduction	3
2	Software Requirements	3
3	System Requirements	3
4	Installation Procedure	4
4.1	Step 1: Install Windows Subsystem for Linux (WSL2)	4
4.2	Step 2: Install Ubuntu	5
4.3	Step 3: Update Ubuntu	6
4.4	Step 4: Install Required Development Packages	6
4.5	Step 5: Install opp_env	7
4.6	Step 6: Create the Workspace	8
4.7	Step 7: Install Simu5G	8
4.8	Step 8: Launch the Simulation Environment	9
4.9	Step 9: Launch OMNeT++	9
4.10	Step 10: Import the INET Project	10
4.11	Step 11: Import the Simu5G Project	10
4.12	Step 12: Build the Projects	10
4.13	Step 13: Open the SingleMecHost Example	11
4.14	Step 14: Run the Simulation	12
4.15	Step 15: Observe the Simulation	12
5	Conclusion	13

1 Introduction

OMNeT++ is an open-source, modular, component-based discrete-event simulation framework widely used for designing and evaluating communication networks. It provides a flexible environment for developing network simulations through reusable modules and graphical tools.

The INET Framework extends OMNeT++ by providing implementations of standard Internet protocols, network devices, mobility models, routing protocols, and wireless communication technologies. It serves as the networking foundation required for advanced simulation models.

Simu5G is a simulation framework built on top of OMNeT++ and INET that enables realistic modeling of 5G New Radio (NR) and LTE cellular networks. It supports features such as standalone (SA) and non-standalone (NSA) architectures, multi-cell deployments, carrier aggregation, dual connectivity, mobile edge computing (MEC), and user mobility.

This manual presents the complete installation procedure of OMNeT++, INET Framework, and Simu5G using Windows Subsystem for Linux (WSL2) and the `opp_env` package manager. The document also demonstrates how to execute a built-in simulation example to verify that the installation has been completed successfully.

2 Software Requirements

Table 1 lists the software versions used during the installation.

Software	Version
Windows	Windows 11 (64-bit)
Windows Subsystem for Linux	WSL2
Ubuntu	Latest LTS
Python	3.x
Git	Latest
OMNeT++	6.4.0
INET Framework	4.6.0
Simu5G	1.4.3
<code>opp_env</code>	Latest Stable Release

Table 1: Software Versions

3 System Requirements

Before starting the installation, ensure that the following minimum system requirements are satisfied.

- Windows 11 (64-bit)
- At least 8 GB RAM
- Minimum 20 GB free disk space
- Stable internet connection
- Administrator privileges
- Microsoft Store access

4 Installation Procedure

This section describes the step-by-step installation procedure for setting up the OMNeT++, INET Framework, and Simu5G simulation environment on Windows using Windows Subsystem for Linux (WSL2). Follow each step in the given order to ensure a successful installation.

4.1 Step 1: Install Windows Subsystem for Linux (WSL2)

Objective

Windows Subsystem for Linux (WSL2) provides a Linux environment within Windows. OMNeT++, INET, and Simu5G will be installed and executed inside this Ubuntu-based Linux environment.

Procedure

- Open **Windows PowerShell** as Administrator.
- Execute the WSL installation command.
- Restart the computer after installation.
- Verify that WSL2 has been installed successfully.

Command

```
1 wsl --install
```

Restart the system and verify the installation.

```
1 wsl --status
```

Expected Output

The terminal should display information similar to:

```
1 Default Distribution: Ubuntu
2 Default Version: 2
3 Kernel version: x.x.x
```

Note

If WSL is already installed, the command displays the current configuration instead of reinstalling it.

4.2 Step 2: Install Ubuntu

Objective

Ubuntu provides the Linux operating system required to build and execute OMNeT++, INET, and Simu5G.

Procedure

- Open Microsoft Store.
- Search for Ubuntu.
- Install the latest Long-Term Support (LTS) version.
- Launch Ubuntu after installation.
- Create a Linux username and password.

Expected Output

After successful installation, the Ubuntu terminal opens and prompts the user to create a new account.

Note

The username and password created here will be used for all future administrative commands inside Ubuntu.

4.3 Step 3: Update Ubuntu

Objective

Updating Ubuntu ensures that all installed packages are upgraded to the latest stable versions before installing development tools.

Procedure

- Open the Ubuntu terminal.
- Update the package repository.
- Upgrade all installed packages.

Command

```
1 sudo apt update
2 sudo apt upgrade -y
```

Expected Output

Ubuntu downloads the latest package information and upgrades all installed software packages successfully.

Note

Depending on the internet connection, this step may take several minutes.

4.4 Step 4: Install Required Development Packages

Objective

The required development packages provide essential tools such as Git, Python, compilers, and utilities needed for installing and building OMNeT++ and Simu5G.

Procedure

- Install Git.
- Install Python and pip.
- Install GNU build tools.
- Install Curl and Wget.
- Verify that all packages are installed correctly.

Command

```
1 sudo apt install -y \  
2 git \  
3 python3 \  
4 python3-pip \  
5 build-essential \  
6 curl \  
7 wget
```

Verify the installation.

```
1 git --version  
2  
3 python3 --version  
4  
5 pip3 --version
```

Expected Output

The installed versions of Git, Python, and pip should be displayed without any errors.

Note

These tools are required by the `opp_env` package manager and for compiling simulation projects.

4.5 Step 5: Install `opp_env`

Objective

Install the `opp_env` package manager, which automatically downloads and manages compatible versions of OMNeT++, INET, and Simu5G.

Procedure

- Install `opp_env` using pip.
- Add it to the system PATH.
- Verify the installation.

Commands

```
1 pip3 install --user opp-env
2 echo 'export PATH=$HOME/.local/bin:$PATH' >> ~/.bashrc
3 source ~/.bashrc
4 opp_env --version
```

Note

The installed version number confirms that opp_env has been installed successfully.

4.6 Step 6: Create the Workspace

Objective

Create a workspace directory to store OMNeT++, INET, and Simu5G.

Command

```
1 mkdir ~/opp_env
2 cd ~/opp_env
```

4.7 Step 7: Install Simu5G

Objective

Download and install Simu5G along with its compatible versions of OMNeT++ and INET.

Command

```
1 opp_env install simu5g-1.4.3
```

The installation automatically downloads:

- OMNeT++ 6.4.0
- INET Framework 4.6.0
- Simu5G 1.4.3

Note

The installation may take several minutes depending on the internet connection and system performance.

4.8 Step 8: Launch the Simulation Environment

Objective

Load the installed simulation environment before starting OMNeT++.

Command

```
1 opp_env shell
```

The terminal should indicate that the environments for OMNeT++, INET, and Simu5G are ready.

4.9 Step 9: Launch OMNeT++

Objective

Start the OMNeT++ Integrated Development Environment (IDE).

Command

```
1 omnetpp
```

The OMNeT++ IDE opens, allowing projects to be imported, built, and executed.

```

To see all the supported simulation models, run:
opp_env list

To create a new workspace and install the latest version of OMNeT++, run:
mkdir workspace && cd workspace && opp_env init && opp_env install omnetpp-latest && opp_env shell omnetpp-latest

To install and use the latest version of the INET Framework with a single command, run:
opp_env shell --install inet-latest

To run a simulation model directly with the latest version of OMNeT++, run:
opp_env run --install omnetpp-latest -c 'cd $OMNETPP_ROOT/samples/loha;./loha'

Run 'touch ~/.opp_env_intro_disable' to remove this intro.

opp_env@036.1.20768515:~/default_workspace$ opp_env shell simu5g-1.4.3
INFO Using specified projects [simu5g-1.4.3] with effective projects [simu5g-1.4.3, inet-4.6.0, omnetpp-6.4.0] in worksp
ace /home/opp_env
INFO Starting non-isolated shell for projects [simu5g-1.4.3, inet-4.6.0, omnetpp-6.4.0] in workspace /home/opp_env
Environment for 'omnetpp-6.4.0' in directory '/home/opp_env/omnetpp-6.4.0' is ready.
Environment for INET 4.6.0 in directory '/home/opp_env/inet-4.6.0' is ready.
/home/opp_env/simu5g-1.4.3 added to PATH
HINT To build, clean, test or check a project or all projects, use the following commands: 'build_all', 'clean_all', 'te
st_all', 'smoke_test_all', 'check_all'; 'build_simu5g', 'clean_simu5g', 'test_simu5g', 'smoke_test_simu5g', 'check_simu5
g'; 'build_inet', 'clean_inet', 'test_inet', 'smoke_test_inet', 'check_inet'; 'build_omnetpp', 'clean_omnetpp', 'test_om
netpp', 'smoke_test_omnetpp', 'check_omnetpp'. (Use 'declare -f <command>' to check what they do.)
HINT Type omnetpp to start the IDE.
omnetpp-6.4.0+inet-4.6.0+simu5g-1.4.3:~/simu5g-1.4.3$ omnetpp
Starting the OMNeT++ IDE...
omnetpp-6.4.0+inet-4.6.0+simu5g-1.4.3:~/simu5g-1.4.3$

```

4.10 Step 10: Import the INET Project

Objective

Import the INET Framework into the OMNeT++ workspace.

Procedure

- Open **File** → **Import**.
- Select **Existing Projects into Workspace**.
- Browse to the `inet-4.6.0` directory.
- Select the project and click **Finish**.

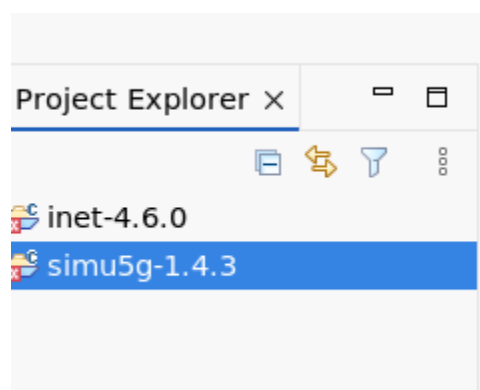
4.11 Step 11: Import the Simu5G Project

Objective

Import the Simu5G project into the OMNeT++ workspace.

Procedure

- Open **File** → **Import**.
- Select **Existing Projects into Workspace**.
- Browse to the `simu5g-1.4.3` directory.
- Select the project and click **Finish**.



4.12 Step 12: Build the Projects

Objective

Compile the imported projects before running any simulation.

Procedure

- Right-click the **INET** project and select **Build Project**.
- Repeat the same for the **Simu5G** project.

Expected Output

The Console should display:

```
1 Build Finished
2
3 0 errors
4 0 warnings
```

4.13 Step 13: Open the SingleMecHost Example

Objective

Open the built-in SingleMecHost simulation to verify the installation.

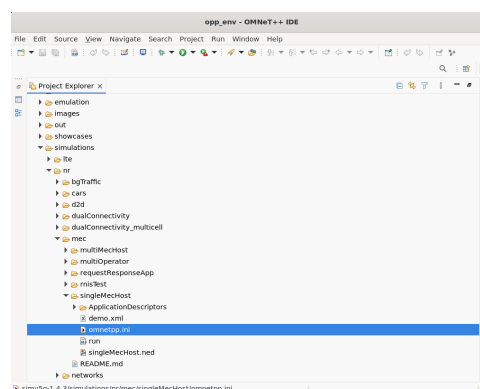
Procedure

Navigate to the following directory in the Project Explorer:

```
1 Simu5G-1.4.3
2     simulations
3         NR
4             Mec
5                 SingleMecHost
```

Open the following files:

- SingleMecHost.ned
- omnetpp.ini



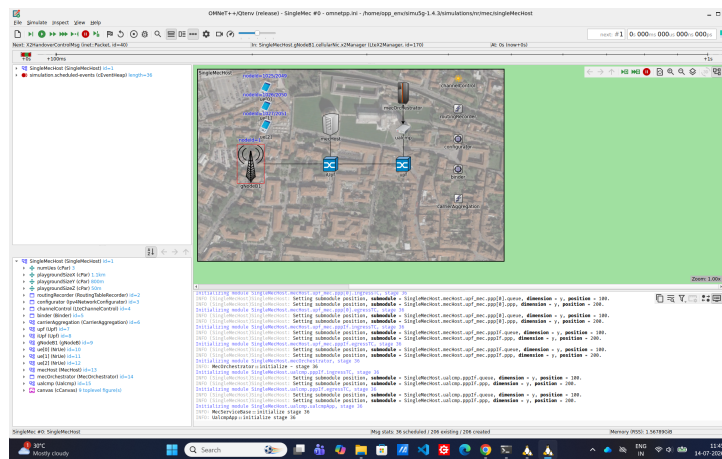
4.14 Step 14: Run the Simulation

Objective

Execute the SingleMecHost simulation using the Qtenv graphical interface.

Procedure

- Right-click `omnetpp.ini`.
- Select **Run As → OMNeT++ Simulation**.
- Choose **Qtenv** if prompted.
- Click **Run**



4.15 Step 15: Observe the Simulation

Objective

Verify that the simulation executes successfully.

Expected Output

During execution, observe:

- UE and gNB communication
- Packet transmission
- Simulation clock
- Event execution

After completion, result files such as `.sca` and `.vec` are generated for analysis.

5 Conclusion

The installation of OMNeT++, INET Framework, and Simu5G was completed successfully using Windows Subsystem for Linux (WSL2) and `opp_env`. The successful execution of the `MyStandAlone` example confirms that the simulation environment is correctly configured and ready for developing and evaluating custom 5G network simulation scenarios.